

Evidence review, architecture decision, and validation plan

A practical path to a faithful Palestinian Arabic lesson transcript that preserves learner errors instead of silently repairing them.

Decision Two synchronized transcript layers	Baseline ElevenLabs timing + targeted OpenAI	Budget \$3 per 60-70 minute lesson
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CRITICAL PRIVACY FINDING

The August 25 lesson transcript is currently reachable on a public GitHub Pages URL with Medi and Amal named. Private storage is a phase-0 requirement.

Prepared for Medi and Amal

4 September 2026 | Research complete | No changes made to C:\dev\anees

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Executive verdict

The project is possible, but "nearly perfect transcript" and "preserve exactly what the learner said" are two different outputs. A conventional speech recognizer is rewarded for producing plausible words. That same behavior can erase the grammatical, lexical, and pronunciation errors Anees needs to find. The right system therefore keeps two synchronized layers:

- 1. Readable/intended transcript - a high-accuracy Arabic/English lesson transcript for review.
- 2. Literal learner record - the closest acoustic account of Medi's actual attempt, including false starts, fillers, omissions, uncertainty, and plausible alternate hearings.

The recommended phase-0 stack is:

- record separate Medi and Amal tracks whenever possible;
- use ElevenLabs Scribe v2 as the current timing/segmentation baseline, because it won Amal's existing blind preference test;
- run OpenAI **gpt-transcribe** on Medi's track or short candidate spans as the literal/verbatim second pass;
- immediately benchmark Munsit and Deepgram Nova-3 **ar-PS** against the same 20 labeled clips per lesson before freezing the clean-transcript provider;
- use Amal's corrections, reformulations, and optional lesson chat as high-value evidence;
- use the personal vocabulary document as the allowed lexical authority by lemma/word family, while quarantining rather than deleting out-of-list forms;
- show Arabic to Amal and Arabizi to Medi from the same timestamped record;
- cap the human review inbox at 20 candidates and a measured 3-5 minutes per lesson.

OpenAI should be part of the solution, but the current Anees evidence does not support making OpenAI the only transcription engine. On the August 25 clips, **gpt-transcribe** avoided a bad diarized-German failure and captured some valuable literal stumbles, but the slice recipe produced 333 words where ElevenLabs produced 420. It retained about 74% as many Medi words and 90% as many Amal words. That makes it a useful second ear, not yet the sole source of truth.

The accepted \$3 budget for a 60-70 minute lesson is sufficient. A production run using one full-file timing transcript plus targeted OpenAI review should normally be well below \$1. A four-provider benchmark can fit below the cap if carefully scoped; Munsit's public credit-to-audio conversion is not documented clearly enough to include in that estimate.

Immediate safety finding

The current repository is public, and the August 25 transcript is live on GitHub Pages with Medi and Amal named. The page returned HTTP 200 during this audit. This conflicts with the sensitivity of private tutoring audio and transcripts. Private storage and authenticated access must precede any automatic pipeline. Removing the already-published transcript requires Medi's explicit approval and is outside this research-only pass.

What "success" means

All three outcomes are first-class requirements:

Outcome	Product definition	Primary measurement
Faithful lesson record	Arabic and English, with timestamps, readable without inventing unspoken content	normalized WER/CER, English-token recall, spot-check accuracy
Correct identity	every turn is Medi or Amal; overlaps and uncertainty are explicit	speaker-attributed WER and speaker-label accuracy

Outcome	Product definition	Primary measurement
Learner-error discovery	preserves and surfaces mistakes, corrections, forgotten-word requests, and uncertainty	error-preservation recall/AWER, candidate precision, review time

The user contract settled during the requirements interview is:

- Palestinian/Levantine Arabic as actually used by Medi and Amal, not Modern Standard Arabic normalization;
- the shared vocabulary document is authoritative, with ordinary inflections, clitics, attached pronouns, plurals, and conjugations allowed by word family;
- English lesson speech is retained; unrelated Farsi or off-call audio is excluded;
- words outside the allowed list are flagged and asked about, never silently accepted or silently rewritten;
- Amal can read Arabic; Medi needs Arabizi;
- Amal usually corrects mistakes, and explicit phrases such as "I forgot the word" or "what does that mean?" are learning events;
- the review batch is 20 clips from each lesson, with heavy setup help acceptable but ongoing review limited to 3-5 minutes;
- pause/retrieval-latency analysis is a later milestone;
- processing budget is \$3 per lesson, separate from recording-software cost.

One detail remains configurable rather than binding: the post-lesson vocabulary grace period. The proposed default is 24 hours, after which the draft is finalized; a later vocabulary update can still re-resolve quarantined tokens.

Research method and evidence quality

This review combined four kinds of evidence:

1. primary product documentation and current pricing for OpenAI, ElevenLabs, Deepgram, Speechmatics, and Munsit;
2. peer-reviewed or shared-task evidence for Palestinian Arabic and learner-speech ASR;
3. open-source implementations that expose reusable annotation and product patterns;
4. Anees's own August 25 experiment files and current source code.

Provider marketing accuracy claims were not treated as transferable benchmarks. Palestinian learner speech, two-party tutoring, code-switching, and deliberate error preservation are a much harder and more specific domain than generic "Arabic" transcription. The Anees gold set is therefore the final decision-maker.

Gap matrix

Requirement	Direct evidence	Best current candidate	Remaining gap
Palestinian clean ASR	NADI 2025 tested 900 Palestinian utterances; the Munsit entry scored 22.27 WER / 8.05 CER	Munsit, then a direct Anees comparison with ElevenLabs and Deepgram	NADI is not Medi/Amal tutoring audio and does not prove learner-error preservation
Exact Palestinian locale	Deepgram documents Nova-3 monolingual ar-PS	Deepgram ar-PS	monolingual mode may lose English code-switches; test ar-PS and language=multi
Arabic-English switching	Munsit exposes munsit-en-ar ; Speechmatics Melia 1 reports Arabic-English results; OpenAI accepts multiple language hints	benchmark all on Anees's mixed clips	no independent head-to-head result on Palestinian learner tutoring
Learner-error preservation	LearnerVoice and ChaLL show standard ASR repairs learner errors and define error-preservation evaluation	targeted OpenAI literal pass now; custom learner model later	published evidence is English, not Palestinian Arabic

Requirement	Direct evidence	Best current candidate	Remaining gap
Word timing	ElevenLabs, Deepgram, Speechmatics, and Munsit expose timing; OpenAI GPT transcription does not expose word timestamps	timing provider + OpenAI on pre-cut spans	must preserve alignment through disagreement and revisions
Medi vs Amal	separate recording channels give deterministic identity	separate tracks; mixed backlog uses diarization plus enrolled voice references	mixed/crosstalk segments remain uncertain and require an explicit "both/unknown" state
Closed personal vocabulary	all major providers have some contextual-bias mechanism; Munsit recommends short 5-30-term lists	dynamic topic/keyterm hints and a post-ASR word-family resolver	the 1,370-word document must be imported, normalized, and tested; never inject all words blindly
Arabic + Arabizi views	Palestinian CODA guidance and Maknuune support normalization/morphology	one canonical record with two renderers	Medi and Amal must settle personal Arabizi conventions through reviewed examples
Error candidates	Amal's corrections and Medi's explicit retrieval questions are strong in-domain signals	evidence-weighted candidate detector	the current tutor-reaction experiment alone is weak (AUC about .52-.57)
3-5 minute review	20-item cap is defined	ranked inbox with audio-first keyboard/phone actions	not yet usability-tested with Amal
Privacy	repository and public page were directly inspected	private object storage + authenticated app	current system fails this requirement

External findings

Palestinian Arabic is still a hard ASR problem

NADI 2025 was the first multidialectal Arabic speech-processing shared task and included a Palestinian test set. The strongest overall entry, Munsit, scored 22.27% Palestinian WER and 8.05% CER. Other entries were roughly 24.87-36.62% WER on Palestinian audio. These numbers matter because they show that generic "98% accuracy" claims do not describe this domain. Character accuracy can be substantially better than word accuracy because Arabic orthography and morphology turn a small character disagreement into a whole-word error.

Munsit is therefore a serious clean-transcript candidate, not an automatic winner. Its standard model accepts hotwords and timestamps; its Arabic-English model supports code-switching but currently ignores hotwords. Its documented batch endpoint is limited to files under 60 minutes, so a 70-minute lesson must be split on silence or streamed. The public plan lists 10,000 free credits and a \$10/month plan with 200,000 credits, but it does not state an auditable speech-to-text credit rate; cost must be measured during the benchmark.

Deepgram added explicit Palestinian Arabic (**ar-PS**) to Nova-3 in January 2026. This is important product evidence, but not a Palestinian learner benchmark. The monolingual and multilingual modes should both be tested: **ar-PS** may be better for Arabic while **multi** is designed for language switching and also supports keyterms.

Speechmatics Melia 1 is a new multilingual model. Speechmatics reports a 15.1% Arabic-English mixed error rate on its own benchmark, ahead of several named competitors, but that is a vendor-run result and does not specify Palestinian learner speech. Melia remains a low-cost challenger, not a settled architectural choice.

ElevenLabs remains the best in-project baseline. Amal's existing blind 20-clip preference test scored the ElevenLabs family 15, Speechmatics 5, and the local dialect model 2. That test measured preferred rendering rather than word-level ground truth, so it should guide the baseline but not end the benchmark.

Standard ASR smooths learner mistakes

The behavior Medi observed is documented in language-learning research. LearnerVoice contains 50.04 hours from 239 one-to-one English tutoring lessons. Its analysis found that stock Whisper disproportionately failed on fillers, self-repairs, and ungrammatical speech and often automatically rectified learner errors. Fine-tuning on learner data reduced overall WER substantially, but the relevant lesson for Anees is structural: a fluent transcript and an error-preserving transcript require different optimization targets.

The ChaLL project goes further by defining Word Annotation Error Rate (AWER) for error preservation. Its annotation scheme distinguishes speech errors, missing words, repetitions, reformulations, long pauses, and uncertain guesses. Its released implementation is English-specific, but its schema and evaluation approach are directly reusable.

There is no API switch that reliably disables all linguistic correction. Prompting OpenAI to preserve fillers, false starts, and mistakes can help, and the OpenAI documentation explicitly recommends prompts for filler preservation and preferred writing systems. It cannot turn a generative recognizer into a raw phonetic instrument. A literal layer must therefore keep alternatives and acoustic evidence rather than force a single confident sentence.

Phone recognition is diagnostic, not a transcript replacement

Allosaurus can produce universal phone sequences and supports more than 2,000 languages. It is worth testing on short, disputed Medi spans to answer questions such as "was there an /r/ here?" It does not know the intended word or Palestinian morphology and is not evidence that a no-language-model system will produce a useful lesson transcript. Its best role is a secondary acoustic diagnostic after the main systems disagree.

Research on Arabic mispronunciation detection also exists, including phone-level work on the L2-KSU dataset. That dataset is read speech with a different variety and task, so its reported accuracy must not be transferred to spontaneous Palestinian tutoring.

Palestinian lexical resources are useful but subordinate

Maknuune provides more than 36,000 Palestinian Arabic entries from 17,000 lemmas and 3,700 roots. Curras provides a revised morphologically annotated Palestinian corpus. Palestinian CODA guidelines describe conventional dialect orthography and explicitly discuss mapping Arabizi to CODA. These resources can expand a user-approved word into plausible inflections, clitics, and spelling variants. They must never override Medi and Amal's own list or silently introduce a different sub-dialect.

What the existing Anees experiments actually say

The repository contains valuable work, but the evidence needs to be interpreted narrowly.

Clean transcript comparison

Amal's blind 20-clip preference result:

Engine family	Preference score
ElevenLabs Scribe v2	15
Speechmatics	5
Local dialect model	2

This supports ElevenLabs as the baseline. It is not WER because the clips do not have a literal word-by-word gold transcript, and Medi has not separately rated how faithfully each engine preserved his attempts.

OpenAI tests

- `gpt-4o-transcribe-diarize` produced a serious language failure on Medi's Arabic in one test, including German-like output.

- **gpt-transcribe** with Arabic and English hints removed that failure and sometimes captured valuable literal stumbles.
- the most promising **gpt-audio-mini** experiment used a model that is now deprecated and cannot anchor production;
- the "slice by ElevenLabs speaker, then transcribe with OpenAI" recipe produced 333 words versus 420 ElevenLabs words across 20 clips: Medi 201/273 and Amal 132/147.

The asymmetry is useful. OpenAI preserved nearly as many Amal words but dropped substantially more Medi words, exactly where hesitations and learner forms are most valuable. It should be evaluated on error-preservation recall, not chosen from a few striking examples.

Tutor-reaction detector

The existing experiment tried to infer errors from what Amal said immediately afterward. Its leave-one-out AUC was about .52-.57 depending on window length-barely above chance. Individual cues such as a short exact repeat had higher precision but low recall. The failure is understandable: mixed-audio diarization sometimes assigns Amal's recast to Medi, and an ASR can already repair Medi's error before the detector sees it.

The result does not mean Amal's corrections are useless. It means reaction text alone is too weak. The new detector must combine channel identity, timing, literal/readable disagreement, explicit learner phrases, Amal's chat, word-family rules, and audio review.

Repository audit

The current code is a useful prototype, not a safe production foundation.

Critical findings

1. Private lesson content is public. **data/lessons/2026-08-25/transcript.txt** and **docs/lessons/2026-08-25.html** are tracked. The pipeline automatically commits and pushes each lesson to a public repository and returns a public Pages link.
2. A failed job is effectively marked processed. The watcher skips every filename already present in **processed.json**, while the exception path writes failed filenames into that same state. A transient API or email failure will not retry automatically.
3. Speaker identity is guessed from Arabic share. The speaker with more Arabic-script tokens is labeled Amal. This can reverse identities when Medi speaks more Arabic, when transcription quality differs by speaker, or when Amal uses English.

High-priority findings

- publication and email are direct side effects of transcription; the pipeline has no draft/approval boundary;
- **git push** and **git commit** failures are ignored, yet the system can still email a link;
- the Gmail sender reads an unrelated project's **.env** through a hard-coded absolute path and borrows that project's **node_modules**;
- the 90-day raw-audio retention rule is documented but not implemented;
- a three-minute Arabic-share sample can incorrectly reject a valid lesson;
- one heuristic equates short tutor confirmations with "Medi said it right," which is not a defensible error label;
- there is no requirements lockfile/package manifest for Python, no isolated email dependency, no CI, and no test suite;
- raw ASR, readable text, literal hypotheses, provenance, confidence, review status, and human corrections are not modeled separately;
- README, task graph, constants, and 960-line blueprint disagree with work already completed and with the current engine evidence.

What is worth keeping

- the reusable word-to-turn structure;
- audio extraction and short-clip generation patterns;

- the existing 20-clip comparison page concept;
- engine adapters as research scripts;
- the green-field idea of a small ranked inbox rather than asking Amal to edit an hour-long transcript;
- the existing learning/practice contracts, after transcription trust is established.

Architecture decision record

Context

Anees must transcribe 60-70 minute Palestinian Arabic/English tutoring sessions, distinguish two known speakers, preserve learner errors, remain within \$3 per lesson, use a personal closed vocabulary, and limit ongoing human review to five minutes. Current input is mixed Google Meet audio; separate-track recording is planned.

Options considered

Option	Advantages	Failure mode
One commercial ASR	simplest and cheapest	clean language model repairs exactly the mistakes Anees needs
One phonetic/CTC model	closer to acoustics	weak readable transcript, timing/diarization burden, no proven Palestinian learner model
Cloud ensemble with two synchronized layers	separates readable accuracy from error preservation; measurable and reversible	more engineering and careful provenance required
Custom model immediately	potentially personal	20 clips are far too little; high risk of overfitting and labeling debt

Decision

Adopt the two-layer cloud ensemble, with separate-track acquisition and human-confirmed gold data. Defer custom fine-tuning until Anees has at least several hundred confirmed error spans and enough acoustic diversity to create honest development and hidden test sets.

Consequences

- no single transcript string is treated as truth;
- every displayed correction links back to a playable audio span;
- provider changes are adapter/config changes rather than data migrations;
- the system can produce a good readable transcript even when the literal layer is uncertain;
- automatic grading remains a later, evidence-gated feature.

Proposed end-to-end system

1. Capture and centralize

For future lessons, use separate local tracks. Zencast is a plausible low-friction option because browser recording can upload each participant's local audio without asking Amal to send a file manually. Its current free-plan feature wording should be verified with a five-minute two-person recording before relying on it for a paid lesson. Google Meet remains the backup.

The central experience should be Anees, not the recording vendor:

- one "New lesson" inbox receives the recording or recording link;
- optional chat text can be pasted or uploaded;
- the lesson remains **draft** during the vocabulary grace period;
- Medi sees Arabizi; Amal sees Arabic; both operate on the same audio/timestamps;
- no public URL is created.

For current mixed recordings, use diarization plus enrolled voice references. Speaker embedding systems such as SpeechBrain ECAPA-TDNN can compare segments with clean Medi and Amal reference clips, but published VoxCeleb results do not guarantee tutoring/crosstalk performance. Low-confidence or overlapping spans must be **both** or **unknown**, not forced to one person.

2. Immutable ingest

Store the source audio privately, hash it, and create an idempotent job keyed by recording ID plus hash. Store each vendor response unchanged with model/version, parameters, time, and cost. Failures remain retryable. No email or publication occurs inside transcription.

3. Segmentation and timing

Use the best timestamped provider on the full lesson or each separate track. ElevenLabs is the initial baseline. Preserve silence intervals and overlaps. Segment on voice activity and sentence-like boundaries; never split through an active word merely to satisfy an API file limit.

4. Transcript hypotheses

For each segment, retain:

- provider transcript(s);
- speaker/channel identity and confidence;
- word or segment timestamps;
- language labels;
- ASR confidence where available;
- a compact list of alternative hearings;
- literal and readable layers as separate fields.

Run OpenAI **gpt-transcribe** on the Medi channel or on 2-12 second spans where there is an error signal, ASR disagreement, low confidence, or an out-of-vocabulary form. Prompt for verbatim Palestinian Arabic, English as English, fillers, repetitions, false starts, and mistakes. Language and keyword hints are allowed; no prompt can be trusted to disable normalization completely.

5. Vocabulary resolver

Import the shared document into versioned entries with:

- canonical lemma/family;
- Arabic spellings;
- Medi's Arabizi spelling(s);
- meaning and notes;
- permitted inflection/clitic rules;
- lesson-added timestamp and reviewer.

Select 5-30 likely rare keyterms dynamically by lesson topic, recent vocabulary, and acoustic candidates. Do not send all 1,370 words to every provider: long hint lists dilute bias and can hallucinate words that were never spoken.

Post-ASR, map recognized forms to allowed families. If no mapping is defensible, keep the audio/text as **unknown_oov** and ask. The resolver may improve the readable layer; it may not overwrite the literal hypothesis.

6. Error/event detector

Create candidates from evidence, not from a language model's confidence alone:

- Amal explicitly negates, repeats, corrects, or reformulates Medi;
- Medi repeats Amal's form after a correction;
- Medi says he forgot a word, asks how to say it, or asks what it means;
- literal and readable hypotheses disagree around Medi's attempt;
- providers disagree or have low acoustic confidence;
- a form is outside the approved vocabulary family;
- Amal's chat provides a spelling/correction near the timestamp;
- later milestone: unusually long retrieval pause relative to Medi's personal baseline.

Assign three review states:

- Confirmed - direct Amal correction/reformulation or authoritative chat evidence;
- Likely - strong acoustic/provider evidence but no explicit correction;
- Check - ambiguous, overlapping, or out-of-list.

An LLM may explain or rank a candidate, but it may not manufacture the learner's original form or declare an error without traceable evidence.

7. Review and output

Rank up to 20 candidates. Each card should play a short context clip and show:

- **What Medi likely said** in Arabizi for Medi;
- **What Medi likely said** in Arabic for Amal;
- **Intended/correct form** in both views;
- why the card was surfaced;
- Confirm / Change / Not an error / New allowed word;
- confidence and alternate hearing only when useful.

Accepted events flow into the existing practice/FSRS plan. Rejected events become negative training/evaluation examples. Every edit preserves the previous value and reviewer.

Benchmark and test strategy

Gold data, 20 clips at a time

For each of the next five lessons, select exactly 20 clips. Across the 100-clip seed set, stratify toward:

- explicit Amal corrections/recasts;
- clean Medi Arabic;
- hesitation, restart, and omission examples;
- Arabic-English switches;
- overlap, low volume, or uncertain speech;
- clean Amal speech.

Medi and Amal label different views of the same record:

- speaker and whether there is overlap;
- literal Medi form (Medi-facing Arabizi and/or Amal-facing Arabic);
- intended/correct form;

- error/event type;
- vocabulary family or new-word status;
- whether the learner form was preserved by each engine.

Provider columns are blinded and randomized. The current Amal preference test remains useful but is not mixed into word-ground-truth metrics.

Provider bake-off

Run the same audio through:

1. ElevenLabs Scribe v2 with timing/diarization and a short keyterm list;
2. Munsit standard and, on code-switch clips, **munsit-en-ar**;
3. Deepgram Nova-3 **ar-PS** and Nova-3 **multi**;
4. OpenAI **gpt-transcribe** with verbatim prompt plus Arabic/English hints;
5. optional challenger: Speechmatics Melia 1 on code-switch clips.

Do not average the two speakers. Report Medi, Amal, code-switch, correction, hesitation, and overlap slices separately.

Metrics

Metric	Why it matters
normalized WER and CER	readable transcript fidelity while controlling spelling variants
speaker-attributed WER / speaker label accuracy	catches correct words assigned to the wrong person
English token recall	prevents Arabic-only tuning from dropping lesson English
filler/restart/omission recall	measures raw learner preservation
AWER or confirmed-error preservation recall	measures whether known errors survive transcription
candidate precision/recall by confidence tier	protects Amal's time and exposes misses
review minutes and edits per 20 cards	validates the product constraint
out-of-vocabulary false alarms	validates the closed-vocabulary resolver
cost and latency per lesson	enforces the budget and operational target

Acceptance targets

These are product gates, not claims that current models already meet them:

- separate-track speaker identity at least 99%; mixed-audio speaker accuracy at least 95% or uncertain spans explicitly withheld;
- Amal clean-speech WER at most 10%; Medi clean-speech WER at most 15%, with CER also reported;
- at least 80% of confirmed learner errors and 80% of fillers/restarts preserved in at least one literal hypothesis;
- Confirmed-tier candidate precision at least 90%; Likely/Check combined precision at least 70%;
- median review time at most five minutes for 20 cards;
- no silent out-of-vocabulary substitution;
- processing cost at most \$3 for a 70-minute lesson.

If the transcript gates miss, Anees can still ship as an audio-first assisted review tool. It must not claim automatic grading.

Test pyramid for the implementation

- Unit tests: word-family expansion, Arabizi normalization, unknown quarantine, state transitions, cost calculation, prompt construction, segment alignment.
- Contract tests: frozen provider response fixtures mapped into a common schema; no live bill during normal CI.
- Integration tests: mixed and separate-track fixture lessons, retry after 429/timeout, crash between stages, vocabulary update during grace period.
- Privacy tests: anonymous/public requests cannot read audio or transcripts; signed links expire; deletion removes all derived clips permitted by policy.
- End-to-end acceptance: one real lesson processed twice produces identical artifacts; Amal completes 20 cards on her phone in under five minutes.
- Regression suite: the hidden fifth-lesson clips plus every previously confirmed failure example.

Cost model for a 70-minute lesson

Current published list prices:

Engine	Published rate	70-minute pass
ElevenLabs Scribe v2	\$0.22/hour	\$0.26
ElevenLabs keyterm add-on	\$0.05/hour	\$0.06
OpenAI gpt-transcribe	\$0.0045/minute	\$0.32
Deepgram Nova-3 monolingual	\$0.0043/minute	\$0.30
Deepgram Nova-3 multilingual	\$0.0052/minute	\$0.36
Speechmatics Melia 1 batch	\$0.129/hour	\$0.15
Speechmatics Enhanced batch	\$0.40/hour	\$0.47
Munsit	credit rate not publicly mapped to audio minutes	measure

The recommended steady-state run is one timestamped full pass plus targeted OpenAI spans, normally about \$0.35-\$0.80 before storage and a small text-model analysis cost. A temporary four-engine benchmark, duplicated tracks, retries, and rounding can still fit a \$1.50-\$2.50 planning band. The application should calculate actual provider-reported usage and stop optional challenger calls before exceeding \$3.

Separate tracks can double billed duration if a vendor charges for full track length including silence. Voice-activity trimming can reduce this, but the budget calculation should assume the conservative double-duration case until measured.

Implementation sequence after approval

Phase 0A - privacy and pipeline safety

1. disable public transcript publication and automatic Git pushes;
2. move lesson artifacts to private authenticated storage;
3. separate transcription, review, notification, and publication stages;
4. introduce retryable/idempotent job state and immutable provider artifacts;
5. isolate dependencies and secrets; add a requirements/lock strategy and tests.

Phase 0B - gold annotation and bake-off

1. create the common segment/gold schema;
2. import the existing 20 clips and ratings without treating preference as ground truth;
3. build the dual Arabic/Arabizi annotation screen;
4. add Munsit and Deepgram adapters;
5. score 20 clips per lesson and publish a private comparison report.

Phase 0C - two-layer transcript

1. keep ElevenLabs as the initial timing baseline;
2. add targeted OpenAI literal hypotheses;
3. import/version the shared vocabulary by family;
4. build unknown quarantine and evidence-based candidates;
5. validate the 20-card/5-minute workflow.

Later milestones

- pause and retrieval-latency modeling using waveform/VAD evidence rather than missing ASR words;
- personal fine-tuning only after several hundred confirmed spans and a frozen hidden test set;
- automatic grading only if error-preservation and precision gates are met;
- practice/FSRS and progress features after transcript trust is established.

Risk register

Risk	Severity	Control
private lessons remain public	critical	remove public pipeline, private storage, explicit deletion plan
clean ASR hides learner error	critical	immutable literal layer, audio-backed alternatives, AWER gate
wrong speaker corrupts correction logic	high	separate channels; enrollment references; unknown/overlap state
vocabulary bias hallucinates approved words	high	5-30 contextual terms, preserve raw response, compare unprompted control
Arabizi conversion changes pronunciation evidence	high	store Arabic/literal source, personal reversible mapping, human review
overfit to one Aug 25 lesson	high	five lessons, stratified clips, hidden holdout, per-condition scores
Amal review grows beyond five minutes	high	hard 20-card cap, confidence ranking, measure and prune signals
vendor/API model changes	medium	common adapter schema, raw response/version capture, regression clips
\$3 cap exceeded by retries/dual tracks	medium	usage ledger, conservative budget, optional calls stop before cap
premature custom model	medium	require label volume and hidden-set improvement before training

Decision summary

- 1. Yes, Anees is feasible as an assisted, evidence-backed learner-error system.
- 2. No, a single "turn off autocorrect" setting does not solve it.
- 3. OpenAI belongs in the literal second-pass role now, not as the unproven sole transcriber.
- 4. ElevenLabs is the in-project baseline; Munsit and Deepgram are the highest-priority challengers.
- 5. Separate tracks are the biggest accuracy improvement for speaker identity and correction extraction.
- 6. The personal vocabulary must constrain interpretation without overwriting the acoustic record.
- 7. The \$3 budget is adequate. Human labeling quality, not API cost, is the scarce resource.
- 8. Privacy remediation is required before the automatic pipeline runs again.

Claim-to-source ledger

Claim used	Source	Evidence class
gpt-transcribe supports prompts, keyword/language hints and costs \$0.0045/minute	developers.openai.com / gpt-transcribe	official product documentation
prompting can preserve fillers; diarized model supports speaker references but not prompts; word timestamps are limited to Whisper in this API guide	developers.openai.com / speech-to-text	official product documentation
OpenAI diarization model	developers.openai.com / gpt-4o-transcribe-diarize	official product documentation
gpt-audio-mini is deprecated	developers.openai.com / gpt-audio-mini	official product documentation
ElevenLabs Scribe pricing and keyterm add-on	elevenlabs.io / api	official pricing
ElevenLabs word timestamps/diarization capability	help.elevenlabs.io / 33053029255697-What-is-Sp...	official product documentation
ElevenLabs keyterm prompting	elevenlabs.io / keyterm-prompting	official product documentation
Deepgram Nova-3 added Palestinian ar-PS	developers.deepgram.com / 27	official changelog
Deepgram pre-recorded rates	deepgram.com / pricing	official pricing
Deepgram multilingual/keyterm support	developers.deepgram.com / keyterm and developers.deepgram.com / 5	official documentation
Speechmatics current rates and \$100 trial credit	www.speechmatics.com / pricing	official pricing
Speechmatics Arabic-English Melia claim	www.speechmatics.com / melia-code-switching-arab...	provider-run benchmark
Munsit models, timing, hotword behavior, and under-60-minute batch limit	docs.munsit.com / get-started and docs.munsit.com / transcribe	official product documentation
Munsit plan headline pricing	munsit.com / home-new	official pricing, incomplete usage mapping

Claim used	Source	Evidence class
NADI task and Palestinian leaderboard	aclanthology.org / 2025.arabicnlp-sharedtask... and nadi.dlnlp.ai / index.html	shared-task paper/leaderboard
Munsit NADI system description	aclanthology.org / 2025.arabicnlp-sharedtask...	peer-reviewed workshop paper
learner ASR repairs errors and fails disproportionately on L2 features	www.isca-archive.org / kim24v_interspeech.pdf	peer-reviewed paper
AWER and error-preserving annotation schema	github.com / chall_e2e_stt and openreview.net / forum	research implementation/paper
Allosaurus universal phone recognition	github.com / allosaurus and arxiv.org / 2002.11800	research implementation/paper
Maknuune size and scope	aclanthology.org / 2022.wanlp-1.13 and github.com / maknuune_lexicon	peer-reviewed paper/repository
Palestinian morphology corpus	aclanthology.org / 2022.lrec-1.82	peer-reviewed paper
Palestinian CODA/Arabizi mapping guidance	www.jarrar.info / HR15.pdf	technical report
ECAPA speaker verification method	github.com / README.md	open-source research implementation
current public repository and live transcript	github.com / anees and thenatanzi.github.io / 2026-08-25.html	direct inspection, 4 Sep 2026

Important limitations

- No new Munsit or Deepgram call was purchased or run in this research pass; their Anees ranking is unresolved.
- The vocabulary document was not fully normalized into a machine-readable lexicon. Its size and authority come from the user's project contract; one visible tab was sampled only.
- The existing 20-clip Amal result is preference data, not a phonetic gold transcript.
- Published learner-error work is mostly English; transfer to Palestinian Arabic is a hypothesis to test.
- Separate-track recording quality and Zencastr free-plan behavior require a real two-person smoke test.
- "Nearly perfect" is a product aspiration. The design makes uncertainty inspectable and reviewable; it does not claim automatic certainty where the audio is ambiguous.